IMAGE CAPTIONING PROJECT

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Dataset

The Flickr 8K dataset was developed as a new benchmark "for sentence-based image description and search" by Hodosh et. al. in *Framing image description as a ranking task: Data, models and evaluation metrics* (Hodosh et. al.). The dataset consists "of 8,000 images that are each paired with five different captions" that describe the image with "descriptions of the salient entities and events" (Hodosh et. al.). This dataset has two purposes: to aid in image retrieval and to aid in image annotation, for the purposes of this project this dataset will be used for the latter.

Related Work

Visual data captioning has been a heavily studied problem in computer vision, but more so for video rather than images (Vinyals et al.). In *Show and Tell: A Neural Image Caption Generator* by Vinyals et al. the authors developed a Vision Deep CNN which feeds into a Language Generating RNN to annotate images. In this method the CNN serves as the encoder and the RNN serves as the decoder. The authors then tested their novel model on the PASCAL, Flickr 30K, Flickr 8K, and SBU datasets and evaluated via the BLEU score. The authors achieved a BLEU-1 score of 63 on the Flickr 8K dataset, providing a straightforward comparison to this project.

Block Diagrams

See Model in GitHub

Algorithms

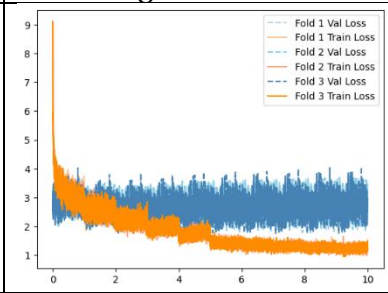
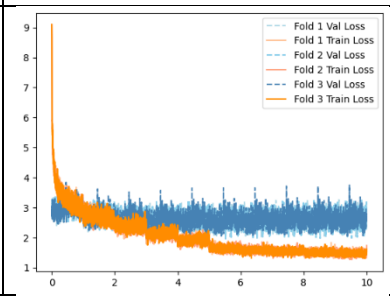
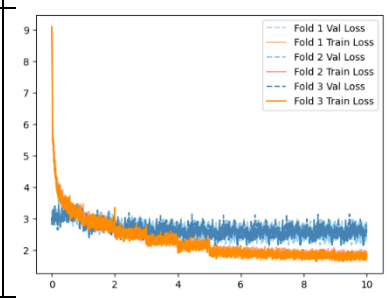
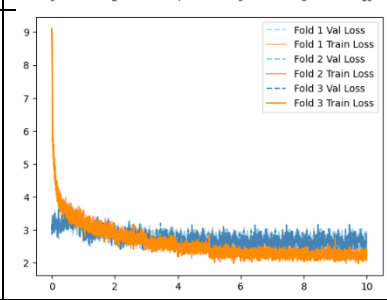
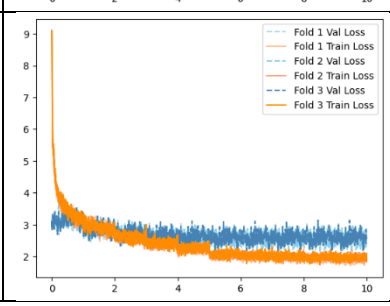
Encoder:

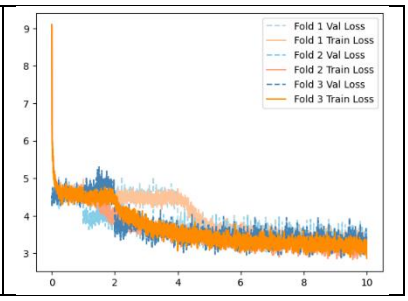
The model employs a pretrained ResNet-152 CNN to extract visual features from the dataset with the removal of the last fully connected layer. Then a linear layer is added to map features to the desired embedding size. Finally, batch normalization and dropout are applied for regularization.

Decoder:

The model employs a multi-layer LSTM (Long Short-Term Memory) network for sequence generation. The model employs an embedding layer to convert word indices to dense vectors and applies dropout for regularization. Then the model concatenates image features with word embeddings as input to the LSTM. Finally, the model employs a linear layer to project LSTM outputs to the vocabulary.

Results

Model	***BLEU-1 Score***	***Training Validation Loss***
Batch Size 16	0.1688	
Batch Size 32	0.1723	
Batch Size 64	0.1726	
Batch Size 64 Dropout 0.5	0.1690	
Batch Size 64 Dropout 0.1	0.1674	

Batch Size 64 Dropout 0.5 Doubled Embedding Size	0.1939	

Comparison and Analysis of Results

To systematically investigate the model's sensitivity and performance, a controlled hyperparameter tuning methodology was implemented, employing a single-variable modification approach to ensure precise attribution of performance variations to specific hyperparameter adjustments. The batch size was selected as the initial hyperparameter for evaluation. A discrete set of batch sizes, specifically 16, 32, and 64, were methodically chosen to explore the performance of the model on the dataset. The dropout layer was subsequently incorporated as the next hyperparameter for evaluation, with dropout rates of 0.1 and 0.5 selected for evaluation. This approach aims to investigate the impact of varying dropout rates on model regularization and performance during training. Finally, the embedding size was increased from 256 to 512. This adjustment is intended to determine whether a larger embedding dimension improves the quality of generated captions.

Overall, the increase in batch size seemed to contribute towards a higher BLEU-1 score. This change may be attributed to the model performing fewer weight updates per epoch, potentially contributing to more stable convergence during training, see the training and validation loss variation in the results table. Similarly, the addition of the dropout layer also contributed towards a higher BLEU-1 score, with higher dropout rates further enhancing this effect. This improvement can be attributed to the regularization effect of dropout, which helps to mitigate overfitting during training. Finally, the increase in embedding size seems to contribute to the highest performance. This enhancement can be attributed to the increased dimensionality of the embeddings, which allows for more complex representations of input features, thereby facilitating better learning of complex patterns. The cross validation did aid in the models not overfitting the dataset. The cross validation seemed to produce a lower training loss over epochs and the validation loss similarly yielded a lower loss over epochs across all models.

While the model with the increased embedding size yielded the highest BLEU-1 score, the model didn't seem to produce the most accurate caption, see Appendix VI. The model with the lowest batch size while not the highest BLEU-1 score did seem to produce an entirely accurate caption for the image, see Appendix I. When testing the model on images outside the dataset, the graduation picture, see Appendix VII, while correctly identified the gender and shirt color, seemed to suggest the city possibly due to the large building. Moreover, the model incorrectly

identified jeans instead of khakis and captioned walking where the image should suggest standing. The beach image, see Appendix VIII, correctly identified the beach scene but incorrectly captioned the nature of the image and incorrectly identified the bird as a dog. The limitations of the Flickr 8K dataset might have contributed towards the misidentification of certain features as mentioned in the previous cases of the images outside of the dataset. The Flickr 30K dataset could possibly contribute towards a better generalization of images outside the dataset with more unseen scenes and objects with annotations in terms of the 8K dataset.

Although, the hyperparameter tuning contributed towards higher BLEU-1 scores. The models still did not outperform the results set by Vinyals et al. Based on the work by He et al. in *Image Captioning through Image Transformer* a model that "uses the original internal transformer layer architecture, with the addition of a gated linear layer on top of the multi-head attention" lead to higher BLEU-1 scores even more so than the Vinyals et al. study (He et al.). For future work, a model that incorporates the transformer architecture seems to outperform the original approach and can contribute towards a higher BLEU-1 score. Moreover, the Flickr 30K dataset might better suit the training of the models to correctly caption images, providing a better generalization of the various scenes and objects, possibly even the images outside the dataset.

References

He, Sen, et al. "Image captioning through image transformer." Proceedings of the Asian conference on computer vision. 2020.

Hodosh, Micah, Peter Young, and Julia Hockenmaier. "Framing image description as a ranking task: Data, models and evaluation metrics." Journal of Artificial Intelligence Research 47 (2013): 853-899.

Vinyals, Oriol, et al. "Show and tell: A neural image caption generator." Proceedings of the IEEE conference on computer vision and pattern recognition. 2015.

Appendices

Appendix I Batch Size 16 - Validation Dataset Image

<start> two dogs are playing together on green grass . <end>

Appendix II Batch Size 32 - Validation Dataset Image

<start> a black dog is running through the water . <end>

Appendix III Batch Size 64 - Validation Dataset Image

<start> a brown dog is running through a field . <end>

Appendix IV Batch Size 64, Dropout 0.5 - Validation Dataset Image

<start> a brown dog is running through the grass . <end>

Appendix V Batch Size 64, Dropout 0.1 - Validation Dataset Image

<start> two dogs are running through a field . <end>

Appendix VI Batch Size 64, Dropout 0.5, Doubled Embedding Size - Validation Dataset Image

<start> a dog is running through the grass . <end>

Appendix VII Batch Size 64, Dropout 0.5, Doubled Embedding Size – Graduation Picture

<start> a man in a white shirt and jeans walks down a street . <end>

Appendix VIII Batch Size 64, Dropout 0.5, Doubled Embedding Size – Beach Picture

<start> a man is standing on a beach with a dog in the water . <end>